

Unit 1: The Real Number System — Practice Worksheet**NC.8.NS.1 · NC.8.NS.2**

Name: _____ Date: _____ Period: _____

Part A — Classify: Rational or Irrational

1. 0.333... (repeating)
(A) Rational (B) Irrational
2. π (3.14159...)
(A) Rational (B) Irrational
3. $\sqrt{49}$
(A) Rational (B) Irrational
4. $\sqrt{10}$
(A) Rational (B) Irrational
5. $-\frac{3}{8}$
(A) Rational (B) Irrational
6. 0.121212... (repeating)
(A) Rational (B) Irrational
7. $\sqrt{144}$
(A) Rational (B) Irrational
8. 0.101001000100001...
(A) Rational (B) Irrational
9. $\frac{2}{7}$
(A) Rational (B) Irrational
10. $\sqrt{2}$
(A) Rational (B) Irrational
11. $\sqrt[3]{64}$
(A) Rational (B) Irrational
12. $\sqrt[3]{30}$
(A) Rational (B) Irrational

Part B — Convert Between Fractions & Decimals

1. Write $\frac{5}{8}$ as a decimal:

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(A) 0.625 (B) 0.85 (C) 0.58 (D) 1.6

2. Write 0.6 as a fraction (simplest form):

(A) $\frac{6}{10}$ (B) $\frac{3}{5}$ (C) $\frac{2}{3}$ (D) $\frac{1}{6}$

3. Write 0.4444... as a fraction (simplest form):

(A) $\frac{4}{10}$ (B) $\frac{2}{5}$ (C) $\frac{4}{9}$ (D) $\frac{1}{4}$

4. Write 0.7272... as a fraction (simplest form):

(A) $\frac{72}{100}$ (B) $\frac{8}{11}$ (C) $\frac{7}{9}$ (D) $\frac{18}{25}$

5. Write $\frac{7}{20}$ as a decimal:

(A) 0.35 (B) 0.72 (C) 0.720 (D) 0.035

6. Write 0.125 as a fraction (simplest form):

(A) $\frac{1}{4}$ (B) $\frac{1}{5}$ (C) $\frac{1}{8}$ (D) $\frac{5}{8}$

Part C — Estimate Each Square or Cube Root

1. $\sqrt{5}$ is closest to:

(A) 2.2 (B) 2.0 (C) 2.5 (D) 2.6

2. $\sqrt{10}$ is closest to:

(A) 3.0 (B) 3.2 (C) 3.6 (D) 3.5

3. $\sqrt{20}$ is closest to:

(A) 4.0 (B) 4.5 (C) 5.0 (D) 4.2

4. $\sqrt{50}$ is closest to:

(A) 7.0 (B) 7.1 (C) 7.5 (D) 6.5

5. $\sqrt{8}$ is closest to:

(A) 2.5 (B) 2.8 (C) 3.0 (D) 2.6

6. $\sqrt{40}$ is closest to:

(A) 6.0 (B) 6.3 (C) 6.5 (D) 7.0

7. $\sqrt[3]{30}$ is closest to:

(A) 3.1 (B) 3.9 (C) 5.5 (D) 2.7

8. $-\sqrt{40}$ is closest to:

(A) 6.3 (B) -6.3 (C) -5.5 (D) -20

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Part D — Order from Least to Greatest

1. $3.5, \sqrt{10}, 10/3, \pi$

2. $1.5, \sqrt{2}, 7/4, \sqrt{3}$

3. $\sqrt{5}, 2.5, 11/5, \sqrt{7}$

4. $-2.5, -\sqrt{5}, -8/3$
